

Student's last name and name:

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Two exercises on the linear theory

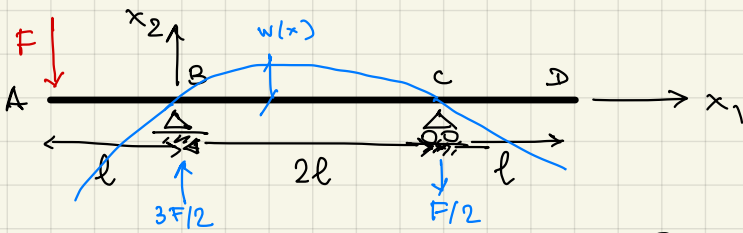
Using linear theory, compute and graph internal forces and displacements. Also, compute reactions and elastic displacements in the cases sketched below, write answers on this sheet

Exercise 1

(total 170 points)

Data: $\underline{F} = -F\mathbf{e}_2$ applied in A

Constraints: $\underline{x}(B) = \underline{0}$
 $\underline{x}(C) \cdot \mathbf{e}_2 = 0$



$$R_{B1} = 0$$

$$R_{B2} = \frac{3F}{2}$$

$$R_{C2} = -\frac{F}{2}$$

(10 points each)

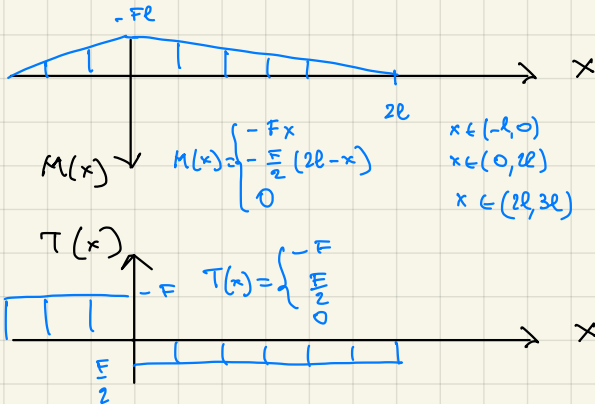
$$\varphi_A = w'(-l) = \frac{7}{6} \frac{Fl^2}{EI}$$

$$\varphi_B = w'(0) = \frac{2}{3} \frac{Fl^2}{EI}$$

$$\varphi_C = w'(2l) = -\frac{Fl^2}{3EI}$$

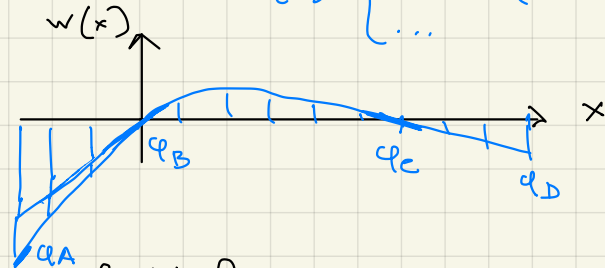
$$\varphi_D = w'(3l) = -\frac{Fl^2}{3EI}$$

(20 points each)



Expressions and diagrams of M, T, w (20 points each)

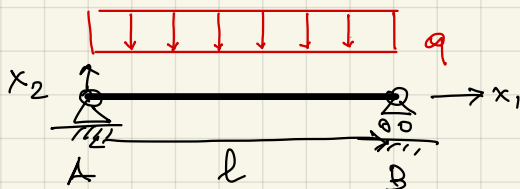
$$w(x) = \begin{cases} \dots \\ \dots \\ \dots \end{cases} \quad (\text{see next page})$$



Exercise 2

(total 120 points)

Using the theorem of virtual power, compute the displacements below



Constraints: $\underline{x}(A) = \underline{0}$

$$\underline{x}(B) \cdot \mathbf{e}_2 = 0$$

Data: $\underline{f} = -q\mathbf{e}_2$

$$\varphi_A = w'(0) = -\frac{1}{24} \frac{ql^3}{EI}$$

$$\varphi_B = w'(l) = \frac{1}{24} \frac{ql^3}{EI}$$

$$w\left(\frac{l}{2}\right) = -\frac{5}{384} \frac{ql^4}{EI}$$

$$u(l) = 0 \quad (30 \text{ points each})$$

$x \in (0, 2\ell)$

$$\frac{M(x)}{EI} = -\frac{F}{2EI} (2l-x) = w''(x) \quad \text{B's} \quad w(0) = 0, w(2l) = 0$$

$$w'(x) = w'(0) - \frac{F}{2EI} \left(2lx - \frac{x^2}{2} \right)$$

$$w(x) = \cancel{w(0)} + w'(0)x - \frac{\pm}{2EI} \left(2\ell \frac{x^2}{2} - \frac{x^3}{6} \right)$$

$$w(2l) = 0 = w'(0) \cdot \frac{2l}{4} - \frac{F}{4EI} \left(\frac{2l}{4} \cdot 4l^2 - \frac{3 \cdot 2l^3}{6} \right) \Rightarrow w'(0) = \frac{F}{4EI} \left(\frac{2l - 8}{6} \right) l^2 = \frac{2}{3} \frac{Fl^2}{EI}$$

$$W(x) = \frac{2}{3} \frac{F l^2}{E I} x - \frac{F}{2 E I} \left(l x^2 - \frac{x^3}{6} \right)$$

$$w'(x) = \frac{2}{3} \frac{F l^2}{E I} - \frac{F}{l E I} \left(2 l x - \frac{x^2}{2} \right)$$

$$w/(2L) = \frac{2}{3} \frac{FL^2}{EI} - \frac{F}{EI} \left(4L^2 - \frac{4L^2}{2} \right) = \frac{FL^2}{EI} \left(\frac{2}{3} - 1 \right) = - \frac{FL^2}{3EI}$$

$$x \in (22, 32)$$

$$\frac{\mu(x)}{\varepsilon} = w''(x) \geq 0 \Rightarrow w'(x) = \text{const.}$$

$$\Rightarrow w'(2\ell) = w'(\ell) = -\frac{7\ell^2}{3\pi J}$$

$$w(x) = \cancel{a} + b(x - 2l)$$

$= 0$ because $w(2l) = 0$

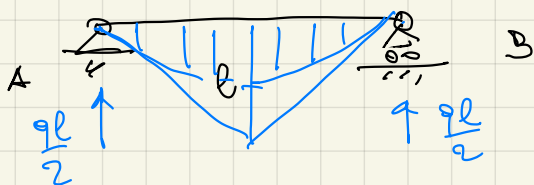
$$b = w'(2l) = - \frac{7l^2}{3EI}$$

$$w(x) = -\frac{Fl^2}{3EI} (x-2l)$$

$x \in (-\ell, 0)$ $\xrightarrow{y=-x}$ $\frac{1}{\mathbb{Z}_I}(y) = -F_y = w''(y)$, $w(0) = 0$, $w'(0) = 0$

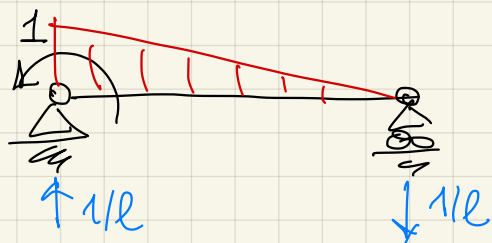
$$w'(y) = \cancel{w'(0)} - F \frac{y^2}{2EI}, \quad w(y) = \cancel{w(0)} - F \frac{y^3}{6EI} \Rightarrow w(x) = \varphi_B x + F \frac{x^3}{6EI}, \quad w'(x) = \varphi_B + \frac{Fx^2}{2EI}$$

$$w^1(-l) = \varphi_A = \frac{Fl^2}{EI} \left(\frac{2}{3} + \frac{1}{2} \right) = \frac{Fl^2}{EI} \cdot \frac{4+3}{6} = \frac{7}{6} \frac{Fl^2}{EI} ; \quad w(x) = \frac{2}{3} \frac{Fl^2}{EI} x + \frac{Fx^3}{6EI}$$



$$M(x) = \frac{q}{2}x - \frac{qx^2}{2} = \frac{q}{2}(lx - x^2)$$

For φ_A

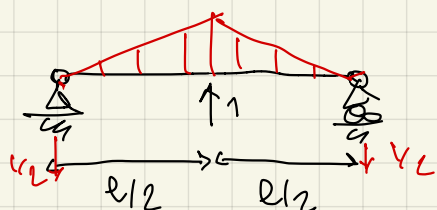


$$M(x) = -\frac{1}{l}(l-x)$$

$$\begin{aligned} \text{PLV: } 1. \varphi_A &= \int_0^l -\frac{1}{l}(l-x) \frac{q}{2EI} (lx - x^2) dx = -\frac{q}{2EI} \int_0^l (l-x)(lx - x^2) dx \\ &= -\frac{q}{2EI} \int_0^l (lx^2 - lx^2 - lx^2 + x^3) dx = -\frac{q}{2EI} \left(\frac{l^4}{2} - \frac{2l^3}{3} + \frac{l^4}{4} \right) \\ &= -\frac{ql^3}{2EI} \frac{6-8+3}{12} = -\frac{ql^3}{24EI} \end{aligned}$$

$\varphi_B = -\varphi_A$ by symmetry

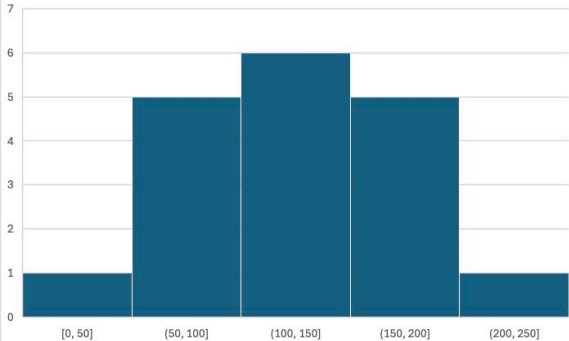
For $w(l/2)$



$$M(x) = \begin{cases} -\frac{1}{2}x & x \in (0, l/2) \\ -\frac{1}{2}(l-x) & x \in (l/2, l) \end{cases}$$

$$\text{PLV: } 1. w(l/2) = \int_0^{l/2} -\frac{1}{2}x \frac{q}{2EI} (lx - x^2) dx + \int_{l/2}^l -\frac{1}{2}(l-x) \frac{q}{2EI} (lx - x^2) dx = -\frac{5}{384} \frac{ql^4}{EI}$$

5/11/2025



28/11/2025

